Gears and wheels

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use gear ratio and wheel circumference to predict an ideal travel speed.

SUCCESS CRITERIA - I can...

1. Connect gear teeth with speed ratio.
2. Calculate distance per wheel revolution.
3. Explain the tradeoff between ideal speed and available torque.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gear ratio” and “circumference” to explain your model or result. Sentence starters are optional.

gear ratio: a comparison of gear tooth counts or rotation rates

circumference: distance around the edge of a circle

torque: the turning effect of a force about an axis

01 Bring back an earlier idea

A motor uses 3 W for 5 s.
Find electrical energy and name a possible non-mechanical output.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $rpm_{out} = rpm_{in} \times teeth_{driver}/teeth_{driven}$; $C = \pi d$
$v = C \times rpm_{wheel}/60$.
Ideal gears and rolling without slip; C in meters gives v in m/s. Loading and losses change real performance.

Worked reasoning: A 10-tooth driver turns a 30-tooth driven gear. The driven gear makes one turn for every three driver turns. In an ideal gear pair, slower rotation trades for greater torque.

Your turn: Draw both gears, label tooth counts and opposite rotation directions.

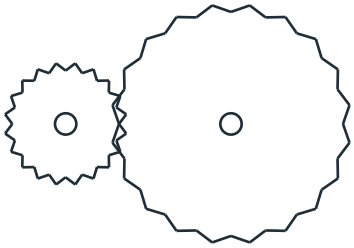
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

A small driver gear turns a larger driven gear.
Predict which turns more slowly and indicate their opposite rotation directions.

Sentence starter: Both … . They differ in … because … .



Driver Driven

Marzano check - circle: ① ② ③

04 Build the solution

Use supplied wheel circumference .20 m and wheel speed 120 rpm.
Find revolutions per second by 120÷60, then ideal speed by .20×that value.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Gears and wheels

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A team chooses the largest wheels because the formula predicts more distance per turn.
Explain why measured acceleration or speed might still decrease.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Choose a method

A total current of 14 A splits between two identical parallel branches.
Find each branch current.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

At the same wheel rpm, doubling circumference does what to ideal travel speed?
State one real-world limitation.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design two different tooth-count pairs giving the same 3:1 reduction.
Explain whether tooth counts alone determine the car's ground speed.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3